# Forward difference under a local derivative variation bound

Define the forward-difference approximation for a real function f at x
with step h as (f(x+h) - f(x)) / h.

Prove that for every f and D from real numbers to real numbers and every
real x, h, L, the following assumptions imply the stated error bound:
- h > 0 and L >= 0;
- f is continuous on the closed interval [x, x+h];
- f has derivative D(t) at every t in the open interval (x, x+h);
- for every t in the open interval (x, x+h),
  |D(t) - D(x)| <= L * |t-x|.

The absolute difference between the defined forward difference and D(x)
is at most L*h. Here D(x) is the reference value in the variation bound.
No derivative at either endpoint is required by this task.